Infinite Series 9

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Problem 5

Let

$$x = \sqrt[p]{k + \sqrt[p]{k + \sqrt[p]{k + \cdots}}}.$$

Then

$$x = \sqrt[p]{k + x},$$
$$x^p - x - k = 0.$$

So x is the root of $x^p - x - k = 0$.

Let $x_1 = \sqrt[p]{k}$, $x_{n+1} = \sqrt[p]{k + x_n}$. Then the sequence x_n converges because it's increasing and bounded. It's increasing because $x_1 = \sqrt[p]{k} < x$, and for any $x_n < x$,

$$x^p - x - k < 0 \implies x^p < k + x \implies x < \sqrt[p]{k + x},$$

so inductively, $x_n < x_{n+1} = \sqrt[p]{k + x}$.

The sequence is bounded because the function $f(x) = \sqrt[p]{k + x}$ is increasing and if $x_n \leq x$ then

$$x_{n+1} = f(x_n) \leq f(x) = x,$$

and $x_1 < x$ so inductively, $x_n \leq x$ for all n.

Problem 6

Let

$$x = \sqrt[n]{q + p\sqrt[n]{q + p\sqrt[n]{q + \cdots}}}.$$

Then

$$x = \sqrt[n]{q + px},$$
$$x^n - px = q,$$

as desired. Now, let

$$y = \sqrt[m/n]{q - p\sqrt[m/n]{q - p\sqrt[m/n]{q - \cdots}}}.$$

Then

$$y^n = \sqrt[m]{q - py}, \quad y^{nm} - py = q.$$

The desired nested radical is equal to y^n, so it satisfies

$$x^m - px^{1/n} = q.$$

Problem 9

Let

$$f(x) = \sqrt{x + \sqrt{x + 1 + \cdots}}.$$

Then

$$f(x) = \sqrt{x + f(x+1)},$$
$$f(x)^2 = x + f(x+1).$$

So the nested radical is equal to $f(1)$ for some f that satisfies the functional equation.

Problem 10

Let

$$f(x) = \sqrt[3]{6x - 2 + (2x-1)^2 \sqrt[3]{(6x+4) + \cdots}}.$$

Then

$$f(x)^3 = 6x - 2 + (2x-1)^2 f(x+1).$$

If f is a polynomial, it must be linear with leading coefficient 2, so we guess that $f(x) = 2x$. The functional equation holds, so the radical is $f(1) = 2$.

Problem 13

Given a cycle of length n, $f^n(x_1) = x_1$, so x_1 is a fixed point of f^n. So stability should be determined using the derivative of f^n. Using the chain rule,

$$(f^n)'(x_1) = f'(x_n) f'(x_{n-1}) \cdots f'(x_1).$$

So stability of a cycle can be defined as: if $|(f^n)'(x_1)| < 1$, the cycle is attracting. If it's > 1, the cycle is repelling. If it's $= 1$, the cycle is neutral.

Problem 15

Let $L = x^{x^{\cdots}}$. Then $L = x^L$, or

$$L = e^{L \log x}.$$

Also,

$$Le^{-L \log x} = 1,$$
$$(-L \log x)e^{-L \log x} = -\log x.$$

By definition of the Lambert W function, $-L \log x = W(-\log x)$, so

$$L = -\frac{W(-\log x)}{\log x}.$$

Problem 16

The power tower converges whenever

$$e^{-e} \leq \frac{1}{x} \leq e^{1/e},$$

so

$$e^{-1/e} \leq x \leq e^e.$$

The value of the power tower is

$$-\left(\frac{W(-\log(1/x))}{\log(1/x)}\right)^{-1} = \frac{\log x}{W(\log x)}.$$

Problem 17

For each part, we test whether

$$W(x)\exp(W(x)) = x.$$

Part (a)

$$\frac{i\pi}{2}e^{i\pi/2} = \frac{i\pi}{2} \cdot i = -\frac{\pi}{2}.$$

Part (b)

$$-1 \cdot e^{-1} = -\frac{1}{e}.$$

Part (c)

$$\log 2 \cdot e^{\log 2} = 2 \log 2.$$